Use the formula $\phi = x/\lambda \times 2\pi$ and $v = f\lambda$

18 A strip of wet cardboard is placed inside a microwave oven. The microwave oven is turned on for a short time. When the card is removed, a pattern of dry spots is observed on the cardboard. This is because a standing wave set up inside the oven.

The dry spots are measured and found to occur at 14 mm, 84 mm, 152 mm, 221 mm and 292 mm from one end of the strip.

From this information, what is the frequency of the microwaves?

A 2.2 GHz **B** 2.6 GHz **C** 4.3 GHz **D** 5.1 GHz

Ans: A

A – A = $1/2\lambda$ = 84 – 14 (A- A represent antinode to anti node distance)

19 Monochromatic light of wavelength 600 nm diffracts through a single slit of width 0.01 mm.